

PROJECT 15 • SDAI TRACK

Workforce Shift Planning & Fatigue Risk Management System

A multi-tenant scheduling platform with rule-based fatigue scoring and an AI explanation layer

TEAM

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DURATION & DOMAIN

- 10-day build sprint
- Domain: workforce safety & scheduling
- Stack: React · Node/Express · MongoDB

Manual shift scheduling is blind to fatigue risk

- **Spreadsheets don't check rest hours.** Managers assign shifts by availability alone — nothing flags that an employee has only 4 hours of rest before their next shift.
- **Overtime stacks invisibly.** Weekly-hour caps and consecutive-day limits are easy to lose track of across dozens of employees and rotating shifts.
- **Fatigue is a safety issue, not a preference.** In healthcare, security, and logistics, fatigued staff drive real incident rates — medical errors, accidents, and near-misses correlate directly with insufficient rest.
- **No audit trail.** When an incident does happen, there's usually no record of whether the schedule that caused it was ever flagged as risky.

WHY IT MATTERS

- Regulatory rest-hour requirements exist across healthcare, transport, and industrial shift work
- Fatigue-related incidents carry direct cost: liability, turnover, lost productivity
- Multi-location organizations can't manually cross-check every employee's rolling 7-day workload
- A system that catches this before publishing the schedule — not after an incident — is the actual product gap

What the system is built on

This isn't a trained ML model — it's a rule engine over a real operational data model, plus an LLM used only for language, never for scoring.

Tech Stack : React with Tailwind for Frontend, Express.js in Backend, Mongo as Database, JWT Token for Authentication etc.

Data model (MongoDB)

- Tenant → Users → Employees
- Shifts (start/end, type, status)
- Availability records
- FatigueRule (per tenant)
- FatigueAssessment (AI output log)

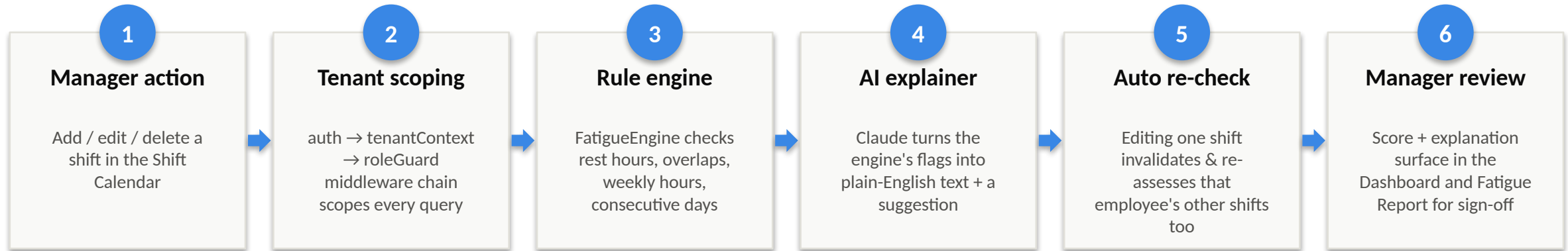
Domain reference rules

- Min rest between shifts: 11h default
- Max consecutive working days: 5
- Max rolling 7-day hours: 48h (or the employee's own stricter cap)
- Risk weight added per violated rule
- Mirrors standard labor-safety rest guidance

Demo / seed dataset

- Seeded demo tenant for testing
- Sample employees across departments
- Realistic shift patterns incl. violations
- Used to validate every risk flag end-to-end
- Same schema every new tenant gets

From a shift edit to a reviewed risk score



ARCHITECTURE

- **MVC + service layer:** routes → middleware → controllers (HTTP only) → services (all business logic) → Mongoose models
- **Multi-tenant by design:** every tenant-scoped collection carries a tenantId; no query anywhere skips it

AI explains risk — it never invents it

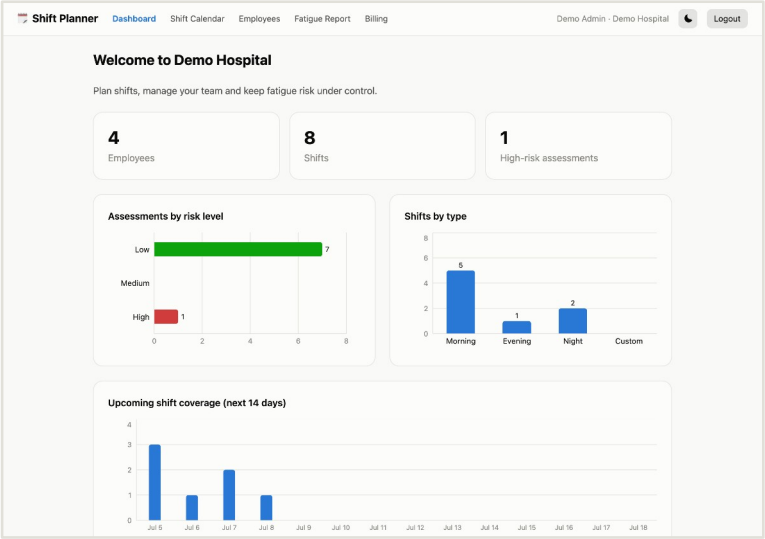
The core innovation: a hard split between deterministic scoring and generative explanation, so a hallucinated number can never reach a manager.

- **Deterministic core:** FatigueEngine computes the risk score from fixed rules — auditable, reproducible, zero AI involvement
- **LLM has one job:** Claude only turns the engine's flags into plain-English text + a suggested alternative
- **Same input every time:** the model only ever sees the engine's own structured flags — never open-ended user text
- **Runs in the background:** assessment never blocks the manager's workflow while it waits on the API

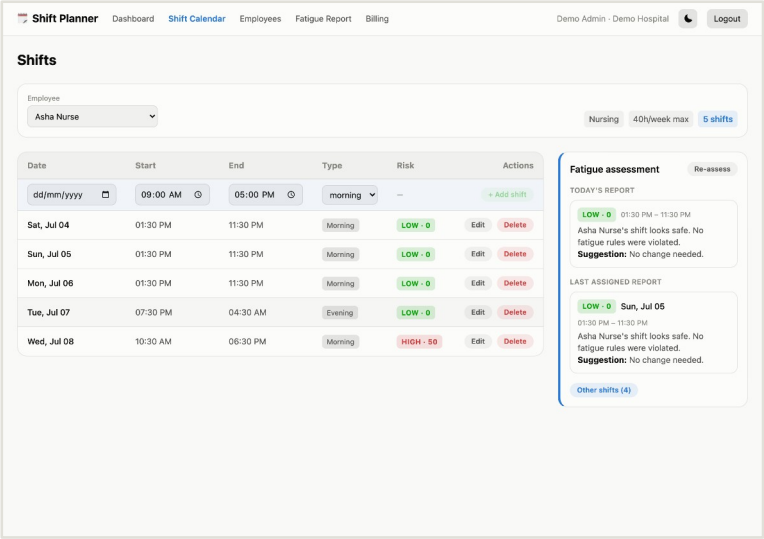
GUARDRAILS IN THE AI LAYER

- **No scoring authority**
AI never sets or influences the risk number
- **Fail-safe fallback**
Falls back to a template if the AI is unavailable
- **Human-in-the-loop**
Every explanation is labeled for manager review
- **Scoped input, no injection surface**
Model sees only engine flags — never raw user text
- **Tenant isolation**
Scoped by the same tenantId as every other query
- **No duplicate / stale results**
Re-assessing a shift replaces its prior result

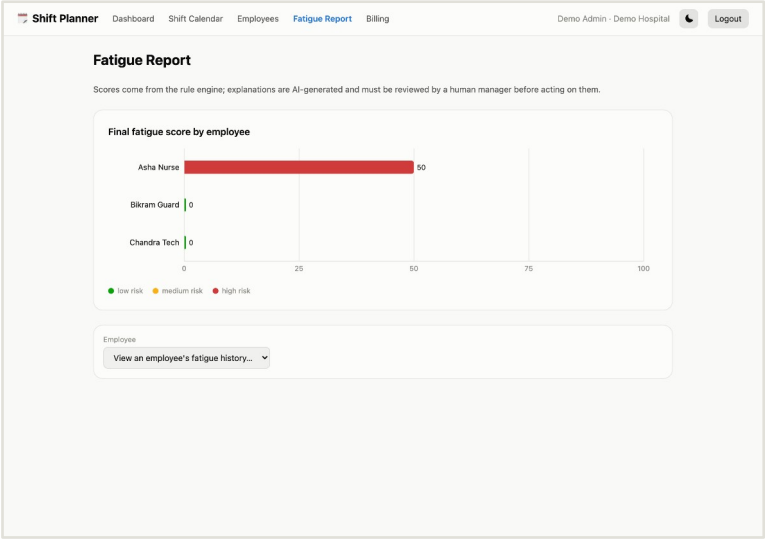
Live product walkthrough



Analytics dashboard



Shift calendar + auto fatigue assessment



Fatigue report & employee risk history

A real assessment produced by the system

HIGH · 50

Asha Nurse — Wed, Jul 08 shift

ENGINE FLAGS (deterministic)

- Only 6.0h rest before this shift (minimum is 11h)
- 47.0h worked in the last 7 days (maximum is 40h)

AI EXPLANATION

"Asha Nurse's shift is rated HIGH risk (score 50/100) because: Only 6.0h rest before this shift (minimum is 11h). 47.0h worked in the last 7 days (maximum is 40h)."

SUGGESTED ALTERNATIVE

Move the shift later, assign it to another available employee, or add a rest day so all rules pass.

DEMO TENANT SNAPSHOT

- **7 low** risk assessments — no rule violations
- **0 medium** risk assessments in this snapshot
- **1 high** risk assessment — flagged automatically, no manual click required
- **8 shifts** across 4 employees, reassessed automatically on every add/edit/delete

Where this system should not be trusted blindly

- **Not a medical or occupational-health judgment.** Rule-based fatigue scoring is a scheduling safeguard, not a substitute for clinical or HR expertise.
- **AI suggestions require sign-off.** Every AI-generated explanation and alternative must be reviewed by a human manager before being acted on.
- **App-layer tenant isolation.** Isolation is enforced by tenantId filtering in code today; a production deployment handling sensitive HR data should add encryption at rest and regular security audits.
- **Demo-mode billing.** Without live payment-gateway keys, Razorpay/PayPal flows run in demo mode — not yet wired to a production billing provider.
- **One fixed rule set per tenant.** Fatigue thresholds are configurable in the data model but have no admin UI yet — a future-improvements item, not a current feature.

Future improvements

Custom rule configuration UI

Let each tenant tune rest-hour minimums, weekly caps, and consecutive-day limits themselves.

Cross-tenant analytics for superadmins

Roll up risk trends across every organization on the platform, not just one tenant at a time.

Push/email alerts on high-risk assignment

Notify a manager the moment a newly created shift is auto-assessed as high risk.

Employee-initiated shift swaps

A request/approval workflow so employees can trade shifts without a manager re-typing the schedule.

AI-layer audit logging

Log every explanation shown to a manager, with rate limiting, as an additional guardrail at scale.

CONCLUSION

Safer schedules, built on rules you can audit — not a black box

- A working multi-tenant SaaS platform, built in 10 days, that turns fatigue risk from a blind spot into a visible, explained, and reviewable number
- Deterministic rule engine for the score, AI strictly for explanation — with guardrails at every step so the model can never hallucinate a safety-critical number
- Automatic, always-current re-assessment: edit one shift, and every affected shift for that employee is quietly re-checked in the background
- Designed to extend: per-tenant rule tuning, cross-tenant analytics, and notification workflows are the clear next steps

Thank you — questions welcome.